

Homework Assignment No. 2

Understanding Consumer Behavior through Discrete Choice Models

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A. Identification of the Study Phenomenon

In the modern digital economy, the final stage of any e-commerce transaction—the check-out—represents a critical juncture for both consumers and merchants. At this moment, consumers must make a discrete choice regarding how to settle their transaction. This study models the **Payment Method Choice** made by consumers on an e-commerce platform.

Understanding consumer behavior in payment method choices has profound practical and theoretical implications:

1. **Financial Optimization for Merchants:** Different payment methods carry distinct transaction fees (merchant discount rates) and processing overheads. By understanding consumer sensitivity to payment characteristics, merchants can design incentives (e.g., discounts for specific methods) to steer users towards lower-cost options.
2. **Checkout Optimization and Conversion Rates:** Transaction speed and security are major determinants of cart abandonment. Dissecting how different demographics value time versus security helps in redesigning the check-out interface to reduce friction.
3. **Product Design & Marketing:** Financial institutions and fintech platforms (like digital wallets) can utilize these behavioral insights to optimize their value propositions, adjusting transaction speeds, security features, or cashback promotions to match target demographic profiles.
4. **Econometric and Behavioral Modeling:** From a discrete choice perspective, this phenomenon represents a classic utility maximization problem where alternatives are mutually exclusive and defined by both alternative-specific attributes (cost, time, security) and individual-specific characteristics (age, gender, membership status).

B. Data Selection and Source

The analysis is based on a comprehensive dataset of an e-commerce platform's transactions. The data is divided into two primary relational tables:

1. **Orders Data (orders.csv):** Captures transaction-level information, including `order_id`, `customer_id`, `total_amount_usd`, the chosen `payment_method`, `order_status`, `delivery_days`, `session_duration_minutes`, `pages_viewed_before_purchase`, and `customer_rating`.
2. **Customers Data (customers.csv):** Captures demographic and historical profile information for each user, including `customer_id`, `country`, `age`, `gender`, `membership_tier` (Free, Silver, Gold, Platinum), and historical purchasing patterns (e.g., `total_spend_usd`, `total_orders`).

To prepare the dataset for discrete choice modeling, we merged the two tables on the common key `customer_id`. The resulting merged dataset contains all necessary components of discrete choice theory:

- **Mutually Exclusive Alternatives (Choice Set):** The set of payment methods available at checkout. We focus on the top 4 options: **Credit Card**, **Debit Card**, **PayPal**, and **UPI / Digital Wallet**.
- **The Choice Variable:** The actual payment method selected by the user for that specific transaction.
- **Alternative-Specific Attributes:**
 - **Transaction Fee (fee_usd):** The direct cost incurred. We calculate this as a percentage of the total order value: Credit Card (2.0%), Debit Card (0.5%), PayPal (3.0%), and UPI / Digital Wallet (0.0%).
 - **Processing Time (processing_time):** Average checkout/settlement duration in seconds: Credit Card (2.0s), Debit Card (2.5s), PayPal (4.0s), and UPI / Digital Wallet (1.5s).
 - **Security Score (security_score):** Perceived security and protection level on a scale from 1 to 10: Credit Card (9.0), Debit Card (8.0), PayPal (9.5), and UPI / Digital Wallet (8.5).
- **Decision-Maker Characteristics:** Individual characteristics merged from the customer profile: `age`, `gender`, and `membership_tier`.

C. Exploratory Data Analysis (EDA)

An exhaustive Exploratory Data Analysis was performed in R to understand the distributions, correlations, and general structures within our sample.

1. Payment Choice Distribution (Market Shares)

Table 1 displays the frequency and market share of the selected payment methods within the cleaned sample.

Table 1: Payment Method Choice Distribution (Estimation Sample)

Payment Method	Count	Market Share (%)
Credit Card	8,557	43.04%
Debit Card	4,956	24.93%
PayPal	4,084	20.54%
UPI / Digital Wallet	2,283	11.48%

Credit Cards dominate the market with a share of 43.04%, followed by Debit Cards (24.93%) and PayPal (20.54%). UPI / Digital Wallet is the least chosen method, with an 11.48% market share.

2. Demographics and Order Values Across Choice Groups

To identify if payment choices vary systematically with individual characteristics, we summarize customer profiles by chosen payment method in Table 2.

Table 2: Averages of Individual Characteristics by Chosen Payment Method

Payment Method	Mean Age	Pct. Female	Mean Order (USD)	Session Duration (min)	Pages Viewed
Credit Card	35.41	48.72%	\$126.26	16.75	6.55
Debit Card	35.67	48.20%	\$127.34	17.08	6.37
PayPal	35.56	50.42%	\$124.90	16.83	6.32
UPI / Wallet	35.67	49.10%	\$127.15	16.65	6.54

A key observation from Table 2 is the striking homogeneity of averages across choice groups. The average age remains virtually constant at around 35.5 years. The gender split is nearly identical, hovering around 48% to 50% female. Crucially, the mean order value is also highly consistent (\$125 to \$127), as are web session durations (~17 minutes) and pages viewed (~6.5 pages). This indicates that, in terms of basic summary statistics, individual covariates do not show strong direct linear relationships with the choice of payment method, which is typical for randomized or balanced synthetic data.

3. Payment Choice by Membership Tier

We also check if loyalty program membership affects payment choice. Table 3 presents the choice probabilities conditioned on the customer's membership tier.

Table 3: Market Shares of Payment Methods by Membership Tier

Membership Tier	Credit Card	Debit Card	PayPal	UPI / Digital Wallet
Free	43.41%	24.38%	20.88%	11.33%
Silver	42.71%	24.92%	20.57%	11.80%
Gold	42.53%	25.74%	19.89%	11.84%
Platinum	42.34%	27.28%	19.37%	11.01%

As membership tier increases from Free to Platinum, there is a very slight increase in the share of Debit Card choices (from 24.38% to 27.28%) and a minor decrease in PayPal choices (from 20.88% to 19.37%). Credit Card and UPI shares remain stable.

4. Representativeness, Missing Values, and Outlier Analysis

- **Representativeness:** The sample represents a large e-commerce user base across multiple countries (with top sales from countries like the USA, Germany, UK, etc.). It represents active users who completed or returned their orders, which is the exact target population for analyzing transaction completion options.
- **Missing Values:** We analyzed missingness across all variables. The only variable containing missing values is `customer_rating` (11,723 missing records). Econometrically, dropping observations due to missing ratings would reduce our estimation sample by more than 58%, introducing potential selection bias. We argue that `customer_rating` is a *post-purchase feedback variable* that occurs after checkout. It does not affect the payment method choice at the time of checkout. Therefore, because it is not a confounder in our choice model, we can safely omit it from the utility equations and preserve the full transaction sample.
- **Outliers:**
 - `total_amount_usd` exhibits a maximum value of \$2,730.88, which is far from the median of \$79.23. A boxplot analysis confirms a long right tail of high-value transactions. However, these are valid large orders on an e-commerce site (not data entry errors). Since transaction fees are linear in order amount, these higher amounts create useful variance in our calculated `fee_usd` attribute.
 - `session_duration_minutes` shows a maximum of 361 minutes. This is realistic for users who leave tabs open during shopping sessions.
 - Customer age ranges from 18 to 75 years, showing a very clean, representative distribution of an adult online shopping population.

D. Definition of the Estimation Sample

To establish a scientifically sound estimation sample for discrete choice models, the original 25,000 transaction records were filtered based on strict econometric criteria:

1. Order Status Filter (Transaction Finalization):

- *Rationale:* We must model the choice made at a finalized checkout. Orders with `order_status` as `Cancelled` (1,456 orders) or `Processing` (1,027 orders) represent incomplete checkouts or transactions aborted before payment settlement was confirmed.
- *Action:* Excluded all orders except those marked as `Delivered` or `Returned`.
- *Sample Size Change:* Reduced from 25,000 to 22,517 observations.

2. Choice Set Definition Filter (Alternative Popularity):

- *Rationale:* Discrete choice models require well-defined choice sets. Minor payment methods like Bank Transfer (832), Buy Now Pay Later (1,358), and Cryptocurrency (447) are chosen in very few transactions, representing less than 12% of total volume combined. Including alternatives with extremely low frequency can cause numerical instability in the estimation of alternative-specific constants and interaction terms.
- *Action:* Restricted the choice set to the four dominant alternatives: Credit Card, Debit Card, PayPal, and UPI / Digital Wallet.
- *Sample Size Change:* Reduced from 22,517 to 19,880 observations.

3. Final Estimation Sample:

- The final estimation sample contains **19,880 clean, transaction-level observations**. This size provides substantial statistical power to estimate the Multinomial Logit (MNL) model parameters with high precision.

E. Choice Model & Econometric Estimation

We formulated and estimated two Multinomial Logit (MNL) models in R using the `mlogit` library.

1. Model Specifications

Model 1: Attribute-Only Model (No ASCs) Model 1 isolates the effect of payment attributes (price, speed, safety) by omitting alternative-specific constants. The utility that individual i derives from alternative j is:

$$U_{ij} = \beta_{price} \cdot fee_usd_{ij} + \beta_{time} \cdot processing_time_j + \beta_{quality} \cdot security_score_j + \gamma_j \cdot X_i + \epsilon_{ij}$$

Here, we interact individual characteristics X_i (age, gender, membership) with the choices. UPI / Digital Wallet is used as the baseline alternative.

Model 2: Full Model with ASCs Model 2 incorporates Alternative-Specific Constants (ASC_j) to capture the baseline preferences for each payment method that are not explained by transaction fees or individual covariates.

$$U_{ij} = ASC_j + \beta_{price} \cdot fee_usd_{ij} + \gamma_j \cdot X_i + \epsilon_{ij}$$

Where $ASC_{CreditCard}$ is normalized to 0.

2. Model Estimation Results

Table 4 summarizes the coefficients estimated for both models.

Table 4: Multinomial Logit Model Estimation Results

Variable	Model 1 (Attribute-Only)		Model 2 (With ASCs)	
	Estimate	p-value	Estimate	p-value
$ASC_{DebitCard}$	–	–	-0.6674	< 2e-16 ***
ASC_{PayPal}	–	–	-0.7311	< 2e-16 ***
ASC_{UPI}	–	–	-1.4153	< 2e-16 ***
Fee (USD) (β_{price})	0.0023	0.618	-0.0038	0.420
Time (s) (β_{time})	-0.2273	< 2e-11 ***	–	–
Security ($\beta_{quality}$)	0.4182	< 2e-14 ***	–	–
Demographic Interactions (vs. Credit Card)				
Age: Debit Card	-0.0007	0.656	0.0020	0.210
Age: PayPal	-0.0097	< 2e-9 ***	0.0011	0.523
Age: UPI	-0.0273	< 2e-16 ***	0.0019	0.358
Male: Debit Card	-0.0028	0.939	0.0184	0.611
Male: PayPal	-0.1567	< 0.0001 ***	-0.0734	0.057 .
Male: UPI	-0.2442	< 2e-7 ***	-0.0238	0.619
Platinum: Debit Card	0.1217	0.067 .	0.1381	0.037 *
Platinum: PayPal	-0.1171	0.111	-0.0496	0.500
Platinum: UPI	-0.1836	0.041 *	-0.0019	0.984
Observations (N)	19,880		19,880	
Log-Likelihood	-25,625.91		-25,492.30	
AIC	51,293.82		51,028.60	
BIC	51,459.67		51,202.34	
Mcfadden R^2	0.0000		0.0004	

3. Econometric Interpretation & Willingness-to-Pay (WTP)

Coefficient Interpretations

- **Time Sensitivity (β_{time}):** In Model 1, the coefficient for processing time is negative and highly statistically significant ($\beta = -0.2273, p < 0.001$). This confirms that consumers prefer faster checkouts, and each additional second of processing time reduces the utility of a payment method.
- **Security Sensitivity ($\beta_{quality}$):** The coefficient for security score is positive and highly significant ($\beta = 0.4182, p < 0.001$). This demonstrates that consumers derive higher utility from payment methods with stronger perceived safety and buyer protections.
- **Price Sensitivity (β_{price}):** In both models, the coefficient for the transaction fee (fee_usd) is statistically insignificant ($p = 0.618$ in Model 1 and $p = 0.420$ in Model 2). Econometrically, we expect price sensitivity to be negative. However, in this dataset, the estimated price coefficients are very close to zero and insignificant.

The Price Insignificance and WTP Challenge From Model 1, we calculated the implied Willingness-to-Pay (WTP): - WTP to save 1 second of processing time: $WTP_{time} = \frac{\beta_{time}}{\beta_{price}} =$

$$\frac{-0.22731}{0.00231} = -\$98.51 \text{ - WTP for 1 point increase in security: } WTP_{security} = \frac{\beta_{quality}}{\beta_{price}} = \frac{0.41818}{0.00231} = \$181.23$$

These results present an interesting econometric challenge: 1. The WTP for transaction speed is negative ($-\$98.51$), which would suggest consumers are willing to pay a premium to *increase* processing time, which is theoretically counterintuitive. 2. The WTP for security is extremely large ($\$181.23$ per point on a 10-point scale), which is economically unrealistic for standard retail checkouts.

The root cause of these unstable and counterintuitive WTP estimates is the **statistical insignificance of the price coefficient** (β_{price}). Because the dataset is synthetic, the simulated choices were generated independently of the actual transaction fees (which are small percentages of the total order value). When estimating the model, the optimizer finds a fee coefficient that is statistically indistinguishable from zero (0.0023 ± 0.0046). Dividing any significant coefficient by a value near zero results in extremely large, volatile, and sign-flipped WTP numbers.

This result highlights a crucial lesson in applied discrete choice modeling: **price sensitivity must be robustly identified and statistically significant** for WTP calculations to carry economic meaning. If price sensitivity cannot be distinguished from zero, the cost variable cannot act as a stable numeraire for translating utility into monetary values.

F. Conclusion

This report successfully completed the groundwork for analyzing consumer payment method choices: 1. We identified a relevant discrete choice phenomenon (Payment Method Choice) and matched it to utility theory. 2. We merged transaction and customer datasets, defined a clean estimation sample of 19,880 observations, and justified the removal of processing/cancelled orders while keeping missing ratings observations. 3. We estimated baseline Multinomial Logit (MNL) models. The model estimation confirmed that transaction speed and security are highly significant drivers of payment choices. 4. We uncovered a critical econometric artifact in the WTP calculation caused by the statistical insignificance of the price coefficient, illustrating the challenges of using weakly identified cost parameters as value numeraires.

This clean sample and these baseline models will serve as the foundation for Homework 3, where we will explore more complex utility specifications, nested structures, and individual heterogeneity.